



LAB MANUAL

NAME:

CMS-ID:

COURSE: DIGITAL LOGIC DESIGN

COURSE SUPERVISOR:

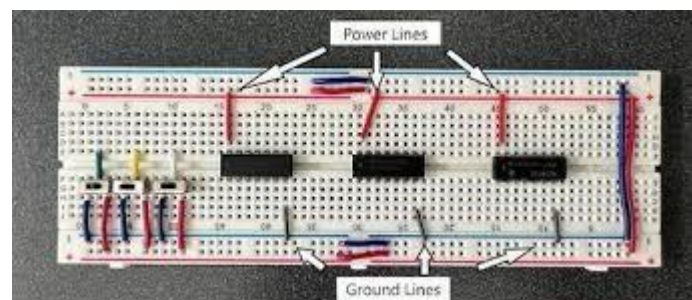
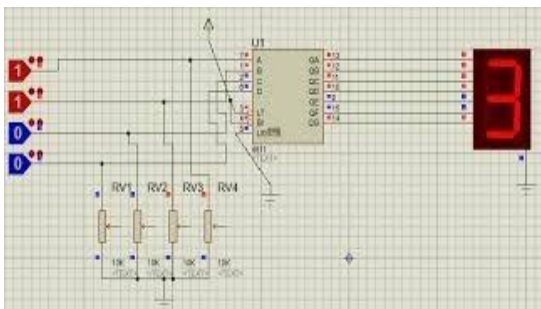
INSTRUCTOR:

COURSE CODE:

SEMESTER: SECOND

PROGRAM:

SECTION:





CERTIFICATE OF COMPLETION

Lab Instructor

Course Supervisor

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LAB 1: INTRODUCTION TO LOGIC GATES (AND, OR, NOT)

Objective:

- Understand the basics of digital logic.
- Learn to implement logic gates in Logisim.
- Verify truth tables of NOT, AND, and OR gates.
- Observe how binary inputs control outputs.

Components Used:

- Logic Gates: AND, OR, NOT
- Input Pins**
- Output LEDs**
- Wires
- Theory and Explanation:
- Digital circuits use binary signals: 0 = LOW, 1 = HIGH.
- AND Gate: Outputs HIGH only when all inputs are HIGH. Logic Expression: $Y = A \cdot B$
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TRUTH TABLE:

A | B | Y

0 | 0 | 0

0 | 1 | 0

1 | 0 | 0

1 | 1 | 1 **OR GATE:** Outputs HIGH if at least one

input is HIGH.

Logic Expression: $Y = A + B$

Truth Table: A | B | Y

0 | 0 | 0

0 | 1 | 1

1 | 0 | 1

1 | 1 | 1

NOT GATE: Outputs the inverse of input.

Logic Expression: $Y = \neg A$

Truth Table: A | Y

0 | 1

1 | 0

These gates are building blocks for all digital circuits.

LAB TASKS:

Open Logisim and create a new project.

Place AND, OR, NOT gates.

Connect input pins and LED outputs.

Test all input combinations.

Observe LED outputs.

Theory Questions:

1. Explain how AND and OR gates differ.
2. What is the purpose of a NOT gate?

Practical Questions:

1. Test AND gate with all input combinations. When does LED glow?

2. Test OR gate with all input combinations.

3. Test NOT gate behavior

LAB 2: UNIVERSAL GATES (NAND AND NOR)

Objective:

Components Used:

- NAND, NOR gates
- Input Pins
- Output LEDs
- Wires

Theory:

1. NAND Gate: Output HIGH except when all inputs are HIGH. Logic:
 $Y = \neg(A \cdot B)$

A **NAND gate** is a **combinational logic gate** that produces an **output HIGH (1)** in all cases **except when all inputs are HIGH (1)**.

2. NOR Gate: Output HIGH only when all inputs are LOW. Logic: $Y = \neg(A + B)$

A **NOR gate** is a **combinational logic gate** that produces an **output HIGH (1)** only when **all inputs are LOW (0)**.

3. Universal gates can implement any logic function.

? **NAND and NOR are universal gates.** A **NOT gate** can be made using a single **NAND** or **NOR** gate

LAB TASKS:

1. Implement NAND and NOR in Logisim.
2. Use NAND/NOR to build NOT, AND, OR gates.
3. Test all input combinations.
4. Verify truth tables.

Theory Questions:

1. How can NAND implement NOT gate?
2. Why are NAND and NOR called universal gates?

Practical Questions:

1. Build AND using NAND gates.
-

2. Build OR using NOR gates.

LAB 3: XOR AND XNOR GATES

Objective:

- Understand XOR and XNOR gates.
- Implement in Logisim.
- Verify truth tables.

Components Used:

- XOR, XNOR gates
- Input Pins
- Output LEDs

Theory:

1. XOR Gate: HIGH output if number of HIGH inputs is odd. Logic: $Y = A \oplus B$

Output is 1 if **both inputs are the same**.

2. XNOR Gate: HIGH output if inputs are equal.

Logic: $Y = \neg(A \oplus B)$

Commonly used in **equality detectors, comparators, and parity generators**.

LAB TASKS:

1. Place XOR and XNOR gates in Logisim.
2. Connect input pins and LEDs.
3. Test all input combinations.
4. Record truth tables.

Theory Questions:

1. Why is XOR called exclusive OR?
2. How can XNOR be used for equality detection?

Practical Questions:

1. Verify XOR outputs for all inputs.
-
-

2. Verify XNOR outputs for all inputs.

LAB 4: HALF ADDER AND FULL ADDER

Objective:

- Implement Half and Full Adders.
- Understand binary addition and carry.

Components Used:

- AND, OR, XOR gates
- Input Pins: A, B, Cin
- Output LEDs: Sum, Carry

Theory:

1. Half Adder: $\text{Sum} = A \oplus B$, $\text{Carry} = A \cdot B$

The **Sum** is obtained using **cascaded XOR gates**.

2. Full Adder: $\text{Sum} = A \oplus B \oplus \text{Cin}$, $\text{Carry} = (A \cdot B) + (B \cdot \text{Cin}) + (A \cdot \text{Cin})$

The **Carry** is obtained using **three AND gates** and **an OR gate**.

A full adder can also be implemented using **two half adders** and **an OR gate**.

LAB TASKS:

1. Build Half Adder in Logisim.
2. Build Full Adder using 2 Half Adders and OR gate.
3. Test all input combinations.
4. Observe LED outputs.

Theory Questions:

1. Difference between Half and Full Adder.
2. Why Full Adder needs carry input?

Practical Questions:

1. Test Half Adder with all combinations.
-
-

2. Test Full Adder for all 8 input combinations.
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LAB 5: HALF SUBTRACTOR AND FULL SUBTRACTOR

Objective:

Implement Half and Full Subtractors.
Understand binary subtraction and borrow.

Components Used:

- AND, OR, XOR, NOT gates
- Input Pins
- Output LEDs: Difference, Borrow

Theory:

1. Half Subtractor: Difference = $A \oplus B$, Borrow = $\neg A \cdot B$

A Half Subtractor handles subtraction of two bits.

2. Full Subtractor: Difference = $A \oplus B \oplus \text{Bin}$, Borrow = $\neg A \cdot B + (\neg A + B) \cdot \text{Bin}$

A Full Subtractor handles subtraction with a borrow from a previous stage, making it suitable for multi-bit binary subtraction.

LAB TASKS:

1. Build Half Subtractor.
2. Build Full Subtractor using Half Subtractors.
3. Test all input combinations.

LAB 6: MULTIPLEXERS (2:1 AND 4:1)

Objective:

- Understand MUX operation.
- Implement 2:1 and 4:1 MUX in Logisim.

Components Used:

- AND, OR, NOT gates
- Input Pins: I0-I3
- Select Pins: S0, S1
- Output LED

Theory:

- 2:1 MUX: 2 inputs, 1 output, 1 select line.

Each term represents **one input being selected**.

- 4:1 MUX: 4 inputs, 1 output, 2 select lines.

Select lines and their complements ensure **only one input is active at a time**.

- Logic Expression: $Y = I0 \cdot S1' \cdot S0' + I1 \cdot S1' \cdot S0 + I2 \cdot S1 \cdot S0' + I3 \cdot S1 \cdot S0$

All selected terms are combined using an **OR gate** to produce the output.

LAB TASKS:

1. Build 2:1 and 4:1 MUX in Logisim.
2. Connect LED to output.
3. Test all input and select combinations.

Theory Questions:

1. How do select lines control output?
2. Why is MUX called a data selector?

Practical Questions:

1. Simulate 4:1 MUX with all inputs 0/1.

2. Observe which input appears at output.

